Sri Lanka Institute of Information Technology

PROJECT REGISTRATION FORM

(This form should be completed and submitted on or before 11.55 PM, Friday 17th July, 2020)

The purpose of this form is to allow final year students of the B.Sc. (Hon) degree program to enlist in the final year project group. Enlisting in a project entails specifying the project title and the details of four members in the group, the internal supervisor (compulsory), external supervisor (may be from the industry) and indicating a brief description of the project. The description of the project entered on this form will not be considered as the formal project proposal. It should however indicate the scope of the project and provide the main potential outcome.

PROJECT TITLE (As per the accepted topic assessment form)	Nana Shilpa (නැණ ශිල්ප): Screening and refinement system to identify the risk of Dyscalculia and Dysgraphia learning disabilities in students.
RESEARCH GROUP **(as per the Topic assessment Form)**	Computational Linguistics
PROJECT NUMBER	(will be assigned by the lecture in charge)

PROJECT GROUP MEMBER DETAILS: (Please start with group leader's details)

	STUDENT NAME	STUDENT NO.	CONTACT NO.	EMAIL ADDRESS
1	Hewapathirana C.D.D	IT17182638	0769235896	it17182638@my.sliit.lk
2	A.D. Kalpani Sandupaba Abeysinghe	IT17180108	0717133001	it17180108@my.sliit.lk
3	Maheshani H.M.P	IT17182324	0771736739	it17182324@my.sliit.lk
4	T.J.P.S. Liyanage	IT17186766	0702007758	it17186766@my.sliit.lk

SUPERVISOR Details

Prof. Samantha Thelijjagoda		
Name	Signature	Date

CO-SUPERVISOR Details (will be assigned by the Supervisor, if necessary)

Ms. Jenny Krishara		
Name	Signature	Date

EXTERNAL SUPERVISOR Details (if any, may be from the industry)

Dr W. M.K. Wanigasinghe	Doctor			
Name	Affiliation	Contact Address	Contact Numbers	Signature/Date

ACCEPTANCE BY CDAP MEMBER

Name	Signature	Date

PROJECT DETAILS

Brief Description of your Research Problem: (extract from the topic assessment form)

Even through there is great technical advancement in all areas, we had identified that there are some topics and complications yet to be provided the solutions and improvements. Especially when it comes to the medical territory there are lot of areas which can be addressed through research projects. Learning disabilities like Dyscalculia and Dysgraphia are can be considered as the area which is yet to be discovered and can deliver better improvements with legitimate research.

In most of the scenarios learning disabilities are not identified correctly and treated properly due to lack of awareness. According to the to the National Center for Learning Disabilities, Dyscalculia and Dysgraphia are included to the five most common types of learning disabilities. Dyscalculia and Dysgraphia learning disability conditions can influence the person's ability to write properly and perform mathematical calculations.

Students who are usually identified as the underperforming students by their parents and teachers can be most likely to be struggling due to those learning disorders. Despite of focusing on the early treatments, students may face discomfort, trauma and resistance to the initial diagnosis process conducted via doctors and hospitals.

Even though there are some Information technology-based solutions in order to identify the Dyscalculia and Dysgraphia learning disabilities, they have not been able to generate accurate results due to lack of definite technical screening methods.

Our aim of this research to deliver a mobile application which helps the students to overcome their disabilities by providing a solution which can identify the disability condition in initial stage with more accurate screening methods.

Description of the Solution: (extract from the topic assessment form)

Through the previously conducted research, it has been identified that in Dyscalculia., there are two types which are speaking and written disorder. Most of the currently available applications does not consider about the mentioned types and does not provide solutions for both above mentioned types separately. Since the larger part of the solutions have not considered the Dyscalculia types, the proposed solution will be containing two separate screening methods to identify the risk of the disability and improvement activities for above mentioned Dyscalculia types.

In the initial medical consultation session, we had identified that many students can overcome these learning disabilities if they are identified in the early stages. Therefore, the solution we are going to deliver is based on the students who are 6 – 10 age range. In the current education system, Sinhala Language is used as the primary language in the majority of the schools. In order to deliver a comprehensive solution, the proposed solution will be identifying the risk of Dysgraphia through the Sinhala characters.
To make the screening process more comfortable and accurate, the user interfaces and overall process of the proposed solution will be built by considering the student mindset and psychological factors. The proposed solution will be process different type of student inputs in order to increase the accuracy of results and also be built on key concepts like machine learning, image processing and deep learning.

The process of designing the screening and improvement activities will be based on the direct medical consultation and the data for the proposed system also will be obtained using direct medical guidance.

As a summary, through the proposed solution we will mainly focus and consider about the untouched areas of those mentioned two learning disabilities. First, identify the risk of having these disabilities in students using programable and active practices and then provide a better solution for the students through a mobile application. We will provide the proposed solution through a Mobile application since almost all the people in the world use smart phones since it is more approachable and convenient platform for the many.

Main expected outcomes of the project: (extract from the topic assessment form)

Dyscalculia and Dysgraphia are incurable diseases which caused in the brain which can be managed with only treatment strategies since there is no cure for those diseases yet. The main purpose of this research is to identify the children with Dyscalculia or Dysgraphia (age between 6 - 10), through various programable and active practices and then provide them activities (training activities to overcome the symptoms) in order to reduce the symptoms of the diseases.

1. Identify Lexical and Verbal Dyscalculia and provide activities to improve the condition.
2. Identify Operational and Practognostic Dyscalculia and provide activities to improve the condition.
3. Identify numeric Dysgraphia through Numbers and provide activities to improve the condition.
4. Identify letter Dysgraphia through Sinhala and English Characters and provide activities to improve the condition.

WORKLOAD ALLOCATION (extract from the topic assessment form)

(Please provide a brief description about the workload allocation)

MEMBER 1	**T.J.P.S. Liyanage – IT17186766**

Identify the risk of Lexical Dyscalculia and Verbal Dyscalculia and provide activities to improve the condition.

Dyscalculia is a biologically based disability centered around the ability to perform numeric symbols and arithmetic operations. There are different forms of Dyscalculia (Practognostic, Operational, Graphical, Lexical, Verbal etc.). Among those forms, we focus on identifying the risk of lexical dyscalculia and verbal dyscalculia in this component.

- Lexical Dyscalculia - Difficulty in reading mathematical symbols, numerals and mathematical expressions or equations.
- Verbal Dyscalculia - Difficulty in naming and understanding the amount of mathematical quantities, numbers, terms, symbols which presented verbally.

To identify Lexical Dyscalculia, we provide a mathematical symbol or numerals graphically through the application interface and getting verbal pronunciation of it form students as a voice input to the application. Those voice inputs compared with the trained data model with noise reduction techniques and predict the risk of Lexical Dyscalculia. To identify Verbal Dyscalculia, we provide a pre-recorded voice clip of a mathematical quantity, numbers or symbol and guide student to select the matching answer among the graphically presented answers. Comparing student selection with given voice content, the application will predict the risk of Verbal Dyscalculia. If any kind of risk detected through these process game-based activities will provide by the application and result are shown as a report.

MEMBER 2 **Hewapathirana C.D.D - IT17182638**

Identify Operational and Practognostic Dyscalculia and provide activities to improve the condition.

- Operational Dyscalculia – difficulty in performing mathematical operations and calculus.
- Practognostic Dyscalculia – difficulty in enumerating, manipulating and comparing real objects and pictures.

This disability affects the ability of the mathematical calculation performance of the student. There are numerous means of testing for dyscalculia. The key focus of this component is to observe how the students attempts a question, look for signs for stress and how they are attempting the math concepts.

- Using the input as mathematical quiz based on the addition, subtraction, division and multiplication. Also, take how long user is taking to complete each question. As well as there are some stages of questions. Based on quiz results by using machine learning algorithms (In here machine learning is used as the data mining technique) predict the probability of dyscalculia level (the application will predict the risk of Dyscalculia).
- Provide a training environment applying reinforcement learning.
- Provide training activities based on the level of the dyscalculia (By providing multisensory learning strategy). Each activity results are saved as a report. Then parents can monitor the student's progress.

MEMBER 3 **Maheshani H.M.P - IT17182324**

Identify the risk of Numeric Dysgraphia and provide activities to improve the condition.

The main purpose of this component is to identify whether a student have a risk to suffer numeric dysgraphia or not and if found, provide remedies through the activities since this disease is incurable and only can be improved the situation of the student by the practices. In this stage, process need to be done as follows.

- Images of numbers written by the students have to be gathered as data and using that data the model has to be trained. When a student is checked through this application for this diagnose, the input data (the written number which is saved as image) has to be gone through the model and then the result will be shown as a probability.
- In here, if found positive for this diagnose, activities will be provided by the application. After each activity, results are shown as a report and parents can view the progress of the student as well.
- For this process, neural networks, machine learning algorithms as well as image processing techniques will be used.

MEMBER 4	**A.D. Kalpani Sandupaba Abeysinghe - IT17180108**

Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition.

This component is used to identify the possibility of the letter level dysgraphia where student experience disability in writing the letters in the provided space, poor handwriting.

The flow of this component is listed below:

- Identify letter level dysgraphia by using the inputs of images of letters Sinhala letters which are written by the students in various scenarios.
- In order to train the data model, data will be collected from different students
- The written letter inputs compared with the trained data model with image processing techniques, machine learning algorithms and predict the risk of letter Dysgraphia.
- Provide a training environment applying reinforcement learning for the students in order to improve the condition based on the results.

DECLARATION

"We declare that the project would involve material prepared by the Group members and that it would not fully or partially incorporate any material prepared by other persons for a fee or free of charge or that it would include material previously submitted by a candidate for a Degree or Diploma in any other University or Institute of Higher Learning and that, to the best of our knowledge and belief, it would not incorporate any material previously published or written by another person in relation to another project except with prior written approval from the supervisor and/or the coordinator of such project and that such unauthorized reproductions will construe offences punishable under the SLIIT Regulations.

We are aware, that if we are found guilty for the above mentioned offences or any project related plagiarism, the SLIIT has right to suspend the project at any time and or to suspend us from the examination and or from the Institution for minimum period of one year".

	STUDENT NAME	STUDENT NO.	SIGNATURE
1	Hewapathirana C.D.D	IT17182638	
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